# FER-DOF Network Figure Explorer

Figure 6 exploration and source guide | Nolan Lab | 20 September 2026

**Prototype; not yet carefully vetted**

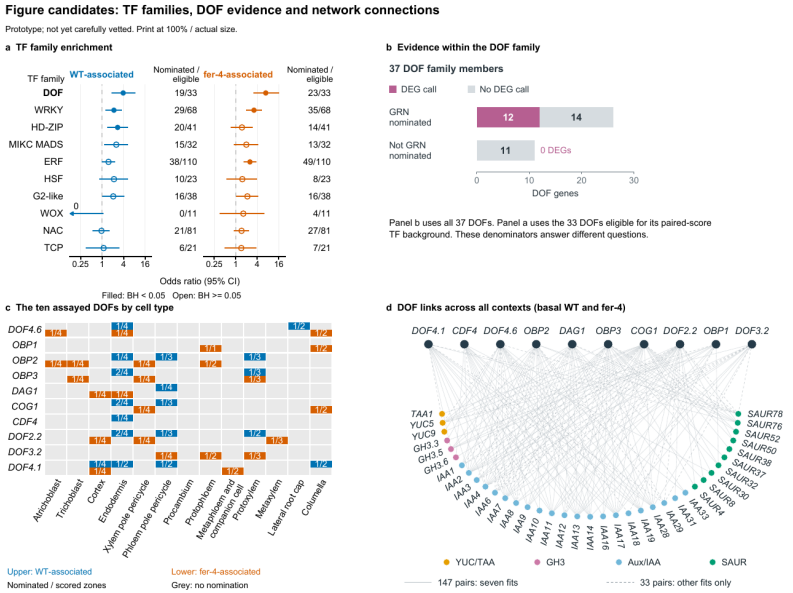
### Start here for Chia-Yu

The following six pages retain the current v7 figure proof and its source appendix, including the DOF-above, auxin-below panel D. Original proof page numbers are preserved.

The package contains plotting code, seven immediate saved-table inputs, editable SVGs, panel PDFs and PNGs, an input/output/code map, and selected audit records.

**What is checked:** saved-table arithmetic and memberships, recorded source-file identities, panel-D edge identity and support classes, and the rendered proof. This packaging step changes navigation only.

**What remains unresolved:** three fer-4 endodermal graph/score mismatches and phloem/xylem gene-feature selection. Source repair and downstream recalculation remain pending. These figures do not establish TF activity or intercellular regulation.




Current four-panel proof; full vector version follows.

**Start-here README and reproduction notes**

**Panel inputs, outputs and code**

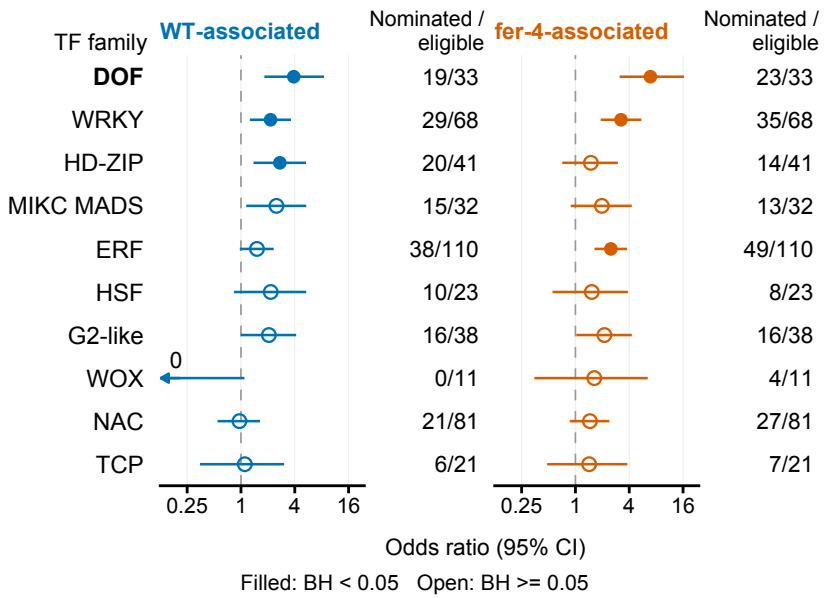
**Validation status and next analytical checks**

Private lab GitLab links require repository access.

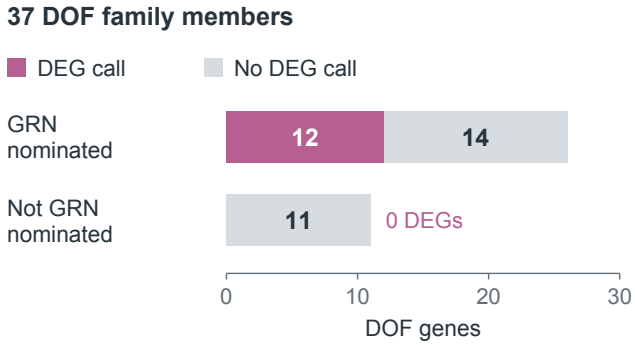
# Figure candidates: TF families, DOF evidence and network connections

Prototype; not yet carefully vetted. Print at 100% / actual size.

## a TF family enrichment

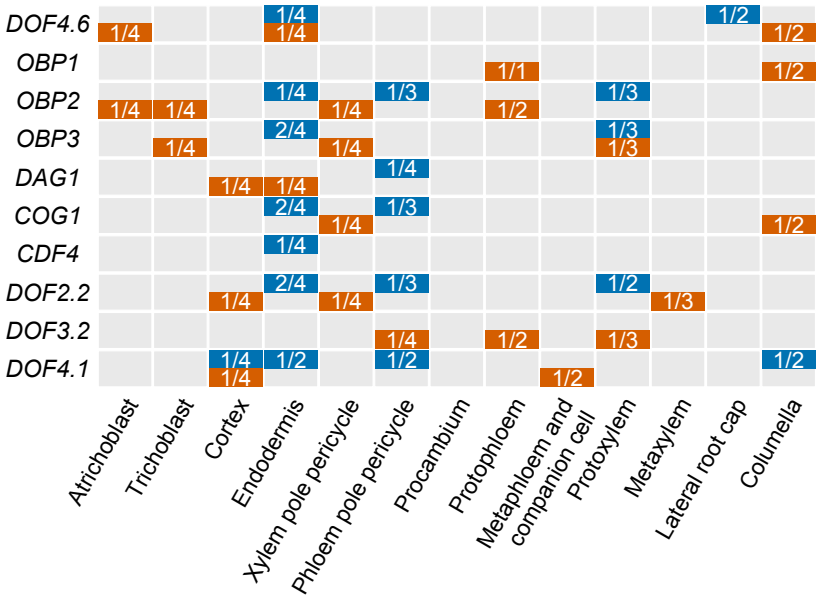



## b Evidence within the DOF family




Panel b uses all 37 DOFs. Panel a uses the 33 DOFs eligible for its paired-score TF background. These denominators answer different questions.

## c The ten assayed DOFs by cell type



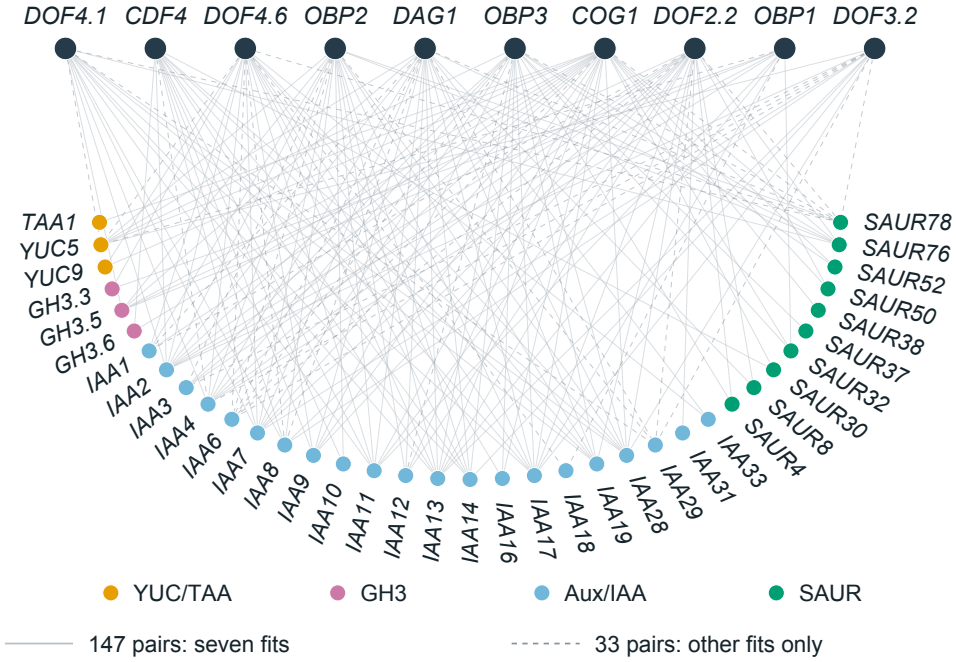

Upper: WT-associated

Lower: fer-4-associated

Nominated / scored zones

Grey: no nomination

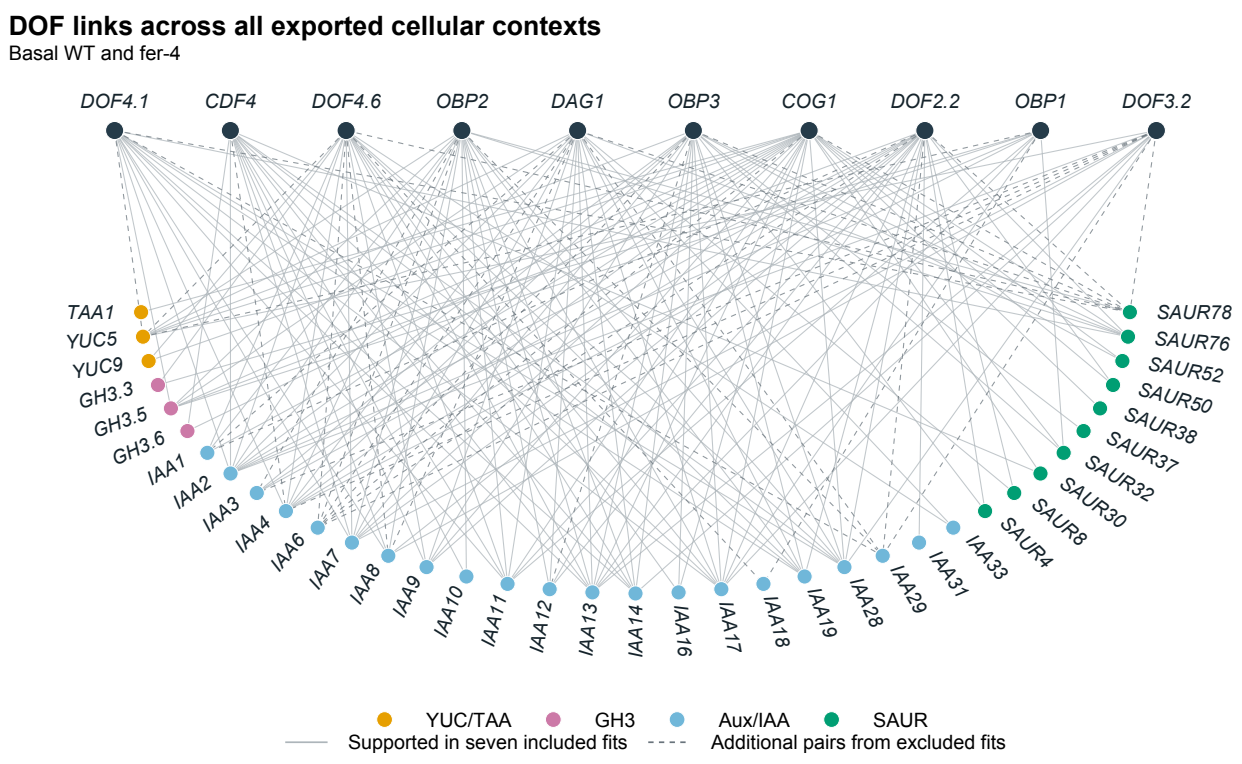
## d DOF links across all contexts (basal WT and fer-4)



# Network detail views

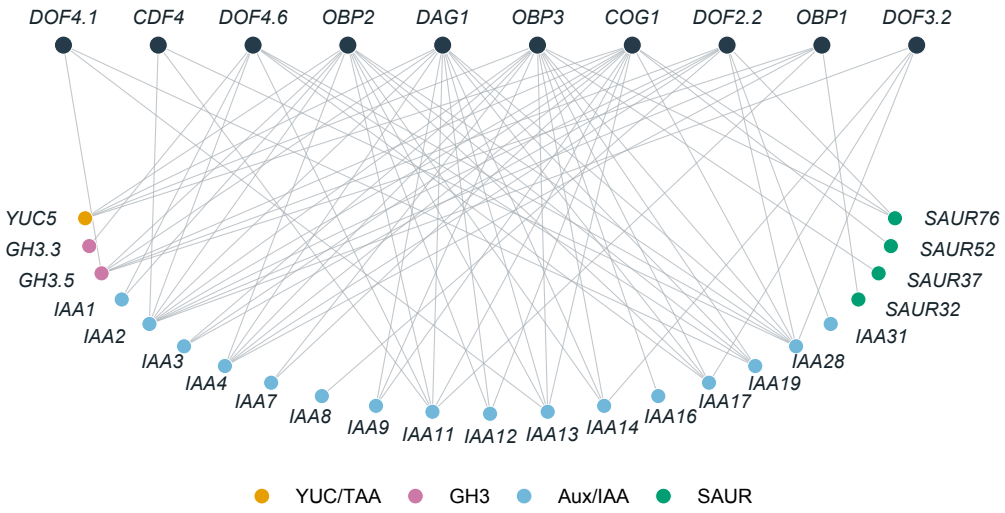
Prototype; not yet carefully vetted. Print at 100% / actual size.

### All-context network, larger view




170 x 100 mm; native vector panel

### Supporting example: elongating pericycle and procambium



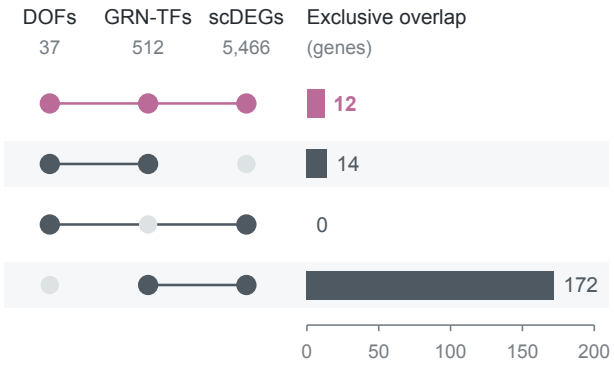

140 x 75 mm; native vector panel

Basal-network unions summarize retained links across contexts; the lines do not represent intercellular transport.

# Retained UpSet option and data provenance

Prototype; not yet carefully vetted. Print at 100% / actual size.

### Alternative b  Exclusive intersections




The inclusive DOF-DEG overlap is 12. All 12 are also GRN-nominated.

The 0 row means DOF + DEG without GRN nomination. Empty dots denote exclusion, not an ignored set.

Singletons omitted from this display: 11 DOFs, 314 GRN-TFs and 5,282 scDEGs.

85 x 55 mm; native vector panel

Underlying data are available. Live checks on 20 September 2026 verified the integrated scRNA-seq object, CellOracle inputs and saved links objects, and full DE tables on Resnick. The paper GitHub repository provides source code and exports; its HEAD matched the pinned revision at the recorded access check.

The limitation is consistency, not missing data. Three saved fer-4 endodermal graphs disagree with their stored degree scores. Separate phloem/xylem target selection has a substring-matching issue. The figure revision does not repair either finding. The independent code and visualization audit is complete and confirms both source issues; neither has been repaired. See the linked source record for exact files and checks.

No model refit or source repair. Independent checks reproduce the saved-table calculations.

# Validation status and source guide

Prototype; not yet carefully vetted. The figures, derived tables and reference code can support discussion and follow-up. They are not final publication results.

The latest presentation revision aligns panels a and c at 110 mm, makes DOF bold, and sets each matrix cell to about 7.1 mm with 8.5 pt numbers and about 1.46 mm side padding. Panel d now occupies the lower-right slot, retaining all 180 pairs with larger nodes and labels. Rendering and arithmetic checks do not establish scientific validity.

**Specific source issues and repair gates**

Endodermis: three fer-4 contexts have filtered graphs that disagree with their stored outgoing-degree scores. First compare the filtered graphs with the existing raw links and declared filter. Recompute scores from the confirmed graph in isolated working directories if needed. A fresh inference fit is not yet shown to be necessary. If scores change, rerun nominations, enrichment and GRN-dependent overlaps.

Phloem and xylem: substring-based target-feature selection admitted genes selected from pole-pericycle DE contexts into the separate tissue fits. This concerns gene features, not accidental inclusion of pericycle cells. Correcting these models requires an explicit map and controlled refitting; deleting edges afterward is not equivalent.

Interpretation: nominations reflect relative network ranks, not measured TF activity. Context unions do not demonstrate co-occurrence within one cell or intercellular transport. YUC5 was excluded before fitting in several tissues, including endodermis. Its missing edges there are not negative biological evidence.

**Audit status**

The requested independent audit report is available: AUDIT_REPORT.md and verification and wording correction. Its scope and unresolved findings must be read before reuse. Audit completion is not publication approval.

Detailed evidence: validation status; source availability; endodermal triage.

**Available data and code**

The pinned paper GitHub repository contains analysis code and exported tables. Integrated scRNA-seq, CellOracle input/raw-link/filtered-link objects and full DE tables are present on Resnick. Exact remote paths, check dates and limits are in the linked source record; large objects were not all reloaded for this layout revision.

The complete input/output/code inventory records exact repository-relative paths, calculation and plotting scripts, outputs, commands and dependencies for all six panel types. Current presentation notes identify the selected panel versions and the new panel d.

The code reference README and source/copy hash manifest accompany 30 reference scripts, each labelled prototype. They are source snapshots with original project dependencies, not a self-contained portable pipeline. Current plotting code is linked on the following pages. No GitLab release has been made.

Additive reproducibility check: membership reconstruction R script and its comparison report. This reproduces memberships from the recorded nomination and DE tables; it does not repair upstream network scores.

# Panel inputs, outputs and code

Links open local files in this checkout. Original source paths and hashes are retained in the CSV inventory. Code snapshots are explicitly marked prototype.

### a: Top-ten TF-family forest | 110 x 80 mm

Inputs: source_top10_family.csv

Calculation: Fisher enrichment from distinct nominated TFs against paired-score Kay TF background; retain PlantTFDB definition; select ten families with >=10 eligible members by max WT/fer4 odds ratio. Preserve 95% two-sided Fisher CI and one-sided enrichment P, BH across 55 families per genotype. Plot existing numbers only.

Plotting code: make_review_v5_forest.R
Output: top10_aligned_forest_review_v5.pdf (PDF; PNG and editable SVG are saved alongside)

Scope: 33 eligible DOFs differs from 37-member family list. Conditional on prior and scored universe; not genotype-difference test. Pooled score nominations inherit unresolved endodermal score/graph and phloem/xylem feature-selection findings.

### b: DOF evidence bars | 85 x 46 mm

Inputs: dot_membership.csv
fig6_full_intersections.csv

Calculation: Count 37 DOFs by nomination and DEG status: 12 both, 14 nominated without DEG, 0 DEG without nomination, 11 neither. Independently assert these four counts against the saved complete intersection table. Stack starting positions are derived from within-group cumulative counts in v6; numerical data remain unchanged.

Plotting code: make_dof_evidence.R
Output: dof_evidence_bars.pdf (PDF; PNG and editable SVG are saved alongside)

Scope: Conditional view of the full 37-member DOF family. DE and network nomination are not independent evidence branches. The original standalone membership generator was not located. A new R reconstruction reproduces the gene identities and flags from the recorded source tables; display-symbol differences are preserved and reported.

### c: Ten-DOF cell-type matrix | 110 x 90 mm

Inputs: celltype_summary_all37.csv

Calculation: Build 37 x 44 gene-context statuses, count WT/fer4 nominations by cell identity, divide by zones scored in both genotypes for the same gene. Select ten assayed DOFs. The v6 plotting script reads only celltype_summary_all37.csv; no TF-family plotting table is read.

Plotting code: make_review_v6_matrix.R
Output: celltype_tested10_review_v6.pdf (PDF; PNG and editable SVG are saved alongside)

Scope: Descriptive context counts, not replication, TF activity, or a cell-type-only fitted network. Opposite directions can occur in different zones. Original pooled score nominations and missingness are preserved, including unresolved source findings.

# Panel inputs, outputs and code (continued)

Links open local files in this checkout. Original source paths and hashes are retained in the CSV inventory. Code snapshots are explicitly marked prototype.

**d: All-context DOF-auxin network | 135 x 90 mm**

Inputs: all_contexts_as_exported_union_edges.csv

Calculation: Read all-ten-tissue tested-DOF edges, normalize target symbols where the annotation table supplies them, select IAA/SAUR/GH3/TAA/TAR/YUC names, collapse source-target pairs across contexts/genotypes. Compare pairs with seven-fit reference. 935 source rows ->180 pairs, 10 DOFs, 37 targets; selected edges in 43 contexts/83 genotype-contexts. 147 pairs have seven-fit support, 33 occur only in excluded fits. V7 changes display geometry, node sizes and typography only.

Plotting code: make_panel_d.R
Output: all_contexts_panel_d.pdf (PDF; PNG and editable SVG are saved alongside)

Scope: As-exported descriptive union, not one jointly fitted graph. Includes all known issue classes; support categories do not establish which edges survive corrected fits. Symbol-family filter is not pathway enrichment or complete auxin biology. Positions are geometry only; no activation, repression or intercellular transport inferred.

**Supporting: Vascular-context network | 140 x 75 mm**

Inputs: tested10_retained_edges_seven_unaffected_consistent_tissues.csv
clean_programme_target_edges.tsv

Calculation: Apply same target-symbol family filter, retain elongating phloem-pole pericycle, xylem-pole pericycle and procambium, and union WT/fer4 source-target pairs. 207 source rows ->81 pairs, 10 DOFs and 23 targets in 3 contexts.

Plotting code: make_overviews.R
Output: vascular_overview_compact.pdf (PDF; PNG and editable SVG are saved alongside)

Scope: Illustrative selected vascular subset. Excludes endodermis/phloem/xylem tissue fits; pole-pericycle contexts are pericycle, not those excluded tissue fits. Union is neither a single fitted graph nor a spatial-transport model. No pooled coefficient or DE encoding.

**Alternative b: Earlier compact UpSet | 85 x 55 mm**

Inputs: fig6_full_intersections.csv

Calculation: Select exclusive multi-set intersections 111,110,101,011 from seven-region table:12,14,0,172. Assert union 5805 and set totals 37,512,5466. Omit singleton bars (11,314,5282), retain complete source table.

Plotting code: make_compact_upset.R
Output: compact_upset.pdf (PDF; PNG and editable SVG are saved alongside)

Scope: Exclusive, not inclusive intersections. DOF+DEG without GRN=0; inclusive DOF-DEG overlap=12. Three singleton regions are omitted from the plotted rows. A new R reconstruction reproduces all seven exclusive counts from the recorded source inputs. Network-dependent counts inherit score-source caveats.